

Carbon Footprint of Bicycle Frame Manufacturing: A Comparative Analysis by Material

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ABSTRACT

Carbon fibre is sold as the premium, lightweight option; by extension, many riders assume it is also the "cleanest". Life-cycle data say otherwise. This document compares the carbon footprint of manufacturing a bicycle frame in five materials —primary aluminium, recycled aluminium, chromoly steel, carbon-fibre-reinforced polymer (CFRP) and titanium— bounding the analysis to *material production* (cradle-to-gate), the stage where most of the impact is decided.

The central result is counterintuitive: frame weight does not predict its footprint. A titanium frame, light, emits on the order of 70 kg of CO₂ in its material; a steel one, heavier, sits around 5–6 kg. The variable that governs is not how much the part weighs but how much energy it costs to produce one kilogram of that material. Recycled aluminium, at roughly 0.6 kg of CO₂ per frame, emits about 95 % less than primary aluminium. And carbon fibre, despite its low weight, lands at the level of primary aluminium and well above steel —with the added problem that today it is essentially non-recyclable.

The practical conclusion for the workshop and the buyer is direct: the choice of material weighs more, environmentally, than any later optimisation; and extending a frame's service life through maintenance is, against the impact of manufacturing it, almost free.

Keywords: *carbon footprint, life-cycle assessment (LCA), embodied energy, bicycle frames, recycled aluminium, carbon fibre, sustainability, emission factor.*

1. The problem and why it is misunderstood

When someone asks "which frame pollutes least?", they are almost always thinking about weight or price. Intuition says the light, expensive option —carbon fibre— must be the most advanced and therefore the cleanest. That intuition is exactly backwards from what the data show, and it is worth understanding why before looking at the numbers.

Producing a material is not free in energy. Mining the ore, refining it, shaping it and delivering it ready to weld or lay up consumes electricity and fuel, and that energy translates into CO₂ emissions. The energy spent up to having the material at the factory gate is called *embodied energy*, and the associated emissions, *embodied carbon*. It is a property of the material, not of the finished product, and it is where most of the impact is decided: in frames, material production dominates the total.

It is worth fixing the boundary of the analysis from the start, because mixing boundaries is the most common cause of figures that seem to contradict each other. Here we speak of the *cradle-to-gate* stage: from ore to formed material. We do not include final transport, use or disposal. When we later cite Trek's figure (a finished carbon frame emits ~57 kg of CO₂), that number is larger because it also covers the frame's manufacturing processes — the lay-up and autoclave curing— not just the material.

2. Embodied energy by material

The first table ranks materials by the energy needed to produce one kilogram, and multiplies it by the typical mass of a frame to obtain the total energy committed to the structure.

$$E_{total} = m_{material} \times E_{embodied} \text{ [MJ]} = [\text{kg}] \times [\text{MJ/kg}]$$

Material	Energy (MJ/kg)	Typical frame (kg)	Total (MJ)
Primary aluminium	180–191	1.5–2.0	270–382
Recycled aluminium	8.1–9	1.5–2.0	12–18
CFRP (carbon fibre)	262–498	1.0–1.3	262–647
Chromoly steel	32	2.2–2.8	70–90
Titanium Ti-3Al-2.5V	~800	1.3–1.6	1040–1280

Compiled from life-cycle inventory databases and LCA literature [1][3][8]. Ranges, not single values.

Recycled aluminium uses about 95 % less energy than primary. The reason is physical: producing primary aluminium requires Bayer-Hall electrolysis, an electrically intensive process; recycling it only requires melting. That leap —from ~180 to ~9 MJ/kg— is the strongest argument for circularity in alloys, and it carries over almost intact to emissions.

3. Carbon footprint by material: why weight misleads

The footprint is obtained by multiplying each material's mass by its emission factor —how many kilograms of CO₂ it costs to produce one kilogram of that material.

$$CO_{2eq} = \sum (m_i \times EF_i) \text{ [EF = emission factor, kg CO}_2\text{/kg]}$$

Material	EF (kg CO ₂ /kg)	Typical frame (kg)	Footprint (kg CO ₂)
Primary aluminium	14.8–18.0	1.8	26.6–32.4
Recycled aluminium	0.26–0.41	1.8	0.47–0.74
CFRP	19.3–34.1	1.15	22.2–39.2
Steel	1.9–2.5	2.5	4.75–6.25
Titanium	~48	1.45	~70

Material production (cradle-to-gate). Compiled from [1][3][8]. The finished-product footprint is larger (see §7).

Here is the point almost nobody explains well. A titanium frame weighs little —about 1.45 kg— and yet it pollutes the most: close to 70 kg of CO₂. A steel frame weighs nearly twice as much and emits ten times less. The paradox vanishes the moment you look at the emission factor: producing one kilo of titanium costs about 48 kg of CO₂; one kilo of steel, barely 2. The weight of the part is a distraction; what decides is the chemistry and the energy of the extraction process. That is why a "light" material can be, by far, the dirtiest to manufacture.

Carbon fibre illustrates the same principle from another angle. Its frame is the lightest (~1.15 kg), but its emission factor is so high that the total footprint (22–39 kg) matches or exceeds primary aluminium. Marketing associates it with technology and lightness; the thermodynamics of its manufacture tells another story.

4. The footprint is amortised through use

A frame emits its footprint once, when manufactured; from then on, every kilometre that replaces a car avoids emissions. The point at which the manufacturing is "paid back" is found by dividing the footprint by the emissions avoided per kilometre.

$$km_{break-even} = Footprint_{frame} \div Emissions_{car} (\approx 0.22 \text{ kg CO}_2/km)$$

Frame material	Footprint (kg CO ₂)	km to offset vs. car
Recycled aluminium	0.6	~3
Steel	5.5	~25
Primary aluminium	30	~136
CFRP	57	~259
Titanium	70	~318

Manufacturing footprint vs. an average car (~220 g CO₂/km) [4][7].

Even the worst case is offset quickly. A finished carbon frame neutralises its manufacture in about 260 km of real use instead of the car —a few weeks for a daily rider. The message is not that the material does not matter, but that the bicycle, as a means of transport, remains extremely low-impact against the car, whatever the frame.

5. Structure vs. consumables: the 142:1 ratio

A common objection is that maintaining a bike —chain, pads, cassette— "also pollutes". It is worth putting this to scale. A cassette's footprint is around 0.4 kg of CO₂; a carbon frame's, 57.

$$R = \text{Footprint}_{\text{frame}} \div \text{Footprint}_{\text{consumable}} = 57 \div 0.4 \approx 142 : 1$$

Replacing wear parts on time generates a negligible impact next to the structure, and in exchange it extends the frame's life by avoiding abnormal loads. Environmentally, maintaining the bike is not a cost: it is what makes the carbon already invested in manufacturing it pay off. The decision that truly matters was made the day the material was chosen.

6. Circularity: the premium-material paradox

Material	Real recyclability	Energy saved by recycling	Status
Aluminium	78–85 %	~95 %	Mature technology
Steel	~85 %	~90 %	Electric-arc furnace
Titanium	~40 %	60–70 %	Limited by contamination
CFRP	< 5 %	—	Downcycling to short fibre only

Compiled from [6][7][8].

Here the correlation one would expect between price and sustainability is inverted. The metals —aluminium and steel— recycle well and return almost all their energy. Carbon fibre, the most expensive and "advanced" material, closes its loop worst: today it is barely recovered through shredding or pyrolysis into low-value short fibre. A carbon frame at end of life is, in practice, waste. That forces a qualification of the word "sustainable" when it accompanies a high-end frame.

7. Reference case: Trek's measured datum

Most of the figures in this document are ranges compiled from literature. A datum measured and published by a manufacturer serves as an anchor. In its 2024 Sustainability Report, Trek quantified the footprint of manufacturing a complete Madone bicycle at **197 kg CO₂e**, of which the frame represents about **29 %**, i.e. **~57 kg CO₂e** [4].

That number is consistent with the above if the boundary is respected: the 22–39 kg in the §3 table are material only; adding the lay-up and autoclave curing of the finished frame brings it to the fifties. The same report confirms that a carbon frame emits on the order of three times more than its aluminium equivalent, closing the loop with the rest of the analysis.

8. Conclusions

Three ideas hold up against the data. First: the material decides. The footprint of manufacturing a frame varies by up to an order of magnitude between recycled aluminium (~0.6 kg) and titanium (~70 kg), and the weight of the part predicts nothing —the material's emission factor does. Second: no material makes the bicycle "dirty"; even the highest-footprint frame is offset within hundreds of kilometres against the car. Third: sustainability does not correlate with price —carbon fibre, premium and light, is among the worst in manufacturing footprint and in recyclability.

For the technician and the buyer, the reading is sober: choosing the material well and then making the frame last through maintenance is, environmentally, what truly moves the needle. The rest is marketing noise.

9. Limitations

This is a synthesis, not a primary LCA. The figures are ranges compiled from inventory databases and literature, with system boundaries that vary between sources; that is why they are presented as intervals and not exact values. The main boundary here is cradle-to-gate (material production); the only finished-product datum is Trek's, flagged as such. Emission factors depend heavily on the electricity mix of the production location —the same material is cleaner made with hydropower than with coal— which adds real spread to the numbers. The qualitative conclusions (the ranking between materials, the dominance of material production, the circularity paradox) are robust; the point values should be read as orders of magnitude.

References

- [1] International Aluminium Institute (2024). *Greenhouse Gas Emissions Intensity — Primary & Recycled Aluminium*. international-aluminium.org
- [2] *Life Cycle Modeling and Carbon Footprint of Bicycle Frames*. Cadernos de Educação, Tecnologia e Sociedade (BRAJETS). brajets.com
- [3] ClimaTiq (2025). *Emission Factors Database — Material Production*. climaTiq.io/data
- [4] Trek Bicycle Corporation (2024). *Sustainability Report 2024*. Madone: 197 kg CO₂e total; frame ≈ 29 % ≈ 57 kg CO₂e. trekbikes.com/sustainability
- [5] American Society for Engineering Education. *Integrating Sustainability Analysis with Design: Case Study of a Bicycle Frame*. peer.asee.org
- [6] PRé Sustainability. *Comparative LCA of Three Materials for Bike Frames*. pre-sustainability.com
- [7] Material Economics (2018). *The Circular Economy — A Powerful Force for Climate Mitigation*. materialeconomics.com
- [8] ecoinvent Association (2024). *ecoinvent v3.10 — Life Cycle Inventory Database*. ecoinvent.org
- [9] ISO 14040:2006 / 14044:2006. *Environmental management — Life Cycle Assessment: principles and framework*. iso.org